

Game Theory: Phishing Emails

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Abstract—To address the common, widespread problem of phishing attacks, especially phishing email attacks on companies, and simulate real-world cybersecurity scenarios, a game theory model was developed. This repeated, dynamic, non-cooperative, and complete-information 2-player game was simulated via a Python program to study player behavior. Eventually, the game converged to a state where the phishing attacker is permanently not attacking, while the defender, or organization, is in a permanent state of low security. The results highlight the importance of the dynamic aspect of the game scenario, where defenders gain more awareness as attackers make more phishing email attempts on the company. The game theory model can be further expanded in the future to better reflect real-world cybersecurity scenarios, such as adding more strategies to each player or dynamic factors.

Index Terms—game theory, phishing, cybersecurity

I. INTRODUCTION

For many people, including individuals and organizations alike, phishing is a widespread cybersecurity issue. Phishing can come in many forms, such as spam emails, URLs, and text messages. Even though some phishing messages targeted at individuals can be obvious to the human eye and intuition, there are more convincing ones that can cause the loss of sensitive information, whether financial or personal. Some groups of people can be especially vulnerable to the risks of phishing, such as the elderly or those not very educated on the problems of cybersecurity. Others can fall victim to it just through the sheer number of phishing messages an individual can receive daily. Through the use of game theory, we address the phishing problem (specifically phishing emails sent to companies) by formulating a game theory model and simulating its behaviors on a Python program.

II. MOTIVATION

Phishing attacks can be especially dangerous for large-scale corporations or organizations, since they can cause greater financial and personal information loss than phishing attacks targeted at individuals. Some of these companies hold sensitive information about not only their employees but also their consumers, especially healthcare or banking organizations. Companies can be susceptible to phishing email attacks when attackers can trick employees into revealing sensitive information. By formulating a game-theory model, we can defend against these attackers by implementing security measures that reduce the likelihood and impact of attacks. The game theory

model can help us better understand the motivation behind the attacker's actions so that companies can improve their defenses.

III. LITERATURE REVIEW

Since phishing is a common problem in the realm of cybersecurity, many papers have addressed it in the past, especially in the context of game theory [2], [4], [5]. Arachchilage and Love [2] developed one of the earlier game theory models to combat phishing attacks, since their paper was published in 2013. Some designed a game theory model for phishing emails specifically [3], [8], [9]. In [3], authors Tchackounte et al. developed a model specifically for spear-fishing emails, where the game scenario was repeated and non-cooperative. Others incorporated their game models with deep learning models [1] or artificial neural networks [6]. In [1], the authors Elberri et al. improved on the CNN deep learning technique by implementing a CNN-LSTM model. Authors Laknho et al. [1] addressed the phishing problem in terms of cryptocurrency exchanges (CCE), since clients could be susceptible to phishing attacks.

A few papers we reviewed addressed phishing by proposing other kinds of solutions rather than the development of a game theory model, such as the development of a serious game to train company employees [14], [15], or by cryptography [18]. Fatima et al. [14] made an empirical evaluation of their game application by conducting a pilot experiment on participants. Kilcullen [18] used his knowledge of game theory to propose an authentication process to protect users from phishing.

IV. SYSTEM MODEL/BACKGROUND

The formulated game theory model is a repeated, dynamic, non-cooperative, and complete-information game with two players: the phishing attacker and the defender, or the organization being attacked. Each player has three strategies to choose from for each game. The phishing attacker can Do Nothing (N), perform Basic Phishing (B), or perform Spear Phishing (S). On the defender's side, the strategies are No Security (NS), Low Security (LS), or High Security (HS).

For the attacker, the utilities include the financial gain from success, the cost of conducting the attack, and the risk of detection or punishment. The equation formulated from these utilities is as follows:

$$U_A(d, a) = P_{\text{success}}(d, a) G(d, a) - P_{\text{failure}}(d, a) F(a) - C_{\text{attack}}(a) \quad (1)$$

In (1), $P_{\text{success}}(d, a)$ is the probability of a successful attack, $P_{\text{failure}}(d, a)$ is the probability of a failed attack, $F(a)$ is the penalty of failing an attack, and $C_{\text{attack}}(a)$ is the cost of executing the attack. $G(d, a)$ is calculated as follows:

$$G(d, a) = \alpha(a) L(d, a) \quad (2)$$

The financial gain $G(d, a)$ is found by multiplying the gain multiplier by the financial loss from successful phishing.

The defender utilities include the cost of maintaining security and the expected financial loss from successful phishing. The following equation is formulated to calculate the utilities:

$$U_D(d, a) = -P_{\text{success}}(d, a) L(d, a) - C_{\text{security}}(d) \quad (3)$$

In (3), the financial loss $L(d, a)$ is calculated by adding four financial loss values, $L_{\text{fraud}}(a)$, $L_{\text{response}}(d)$, $L_{\text{downtime}}(d, a)$, and $L_{\text{reputation}}(a)$.

The value of $P_{\text{success}}(d, a)$ is calculated by multiplying four probability factors: Delivery, Click, Credential, and Monetize. To create a dynamic game scenario, we created a state variable called Defender Awareness that changes depending on the attacker's past actions, and thus influences the value of $P_{\text{success}}(d, a)$. In the Python program used to simulate our dynamic game scenario, this state variable is an integer value ranging from 0 to 5, where it increases by 1 after a phishing attempt has been made and decreases by 1 if the last two rounds had no attacks made. This value cannot be higher than 5 or lower than 0. This variable representing defender awareness is then used to adjust the Click probability factors for the next round of the game, where a higher Awareness factor means a lower $P_{\text{success}}(d, a)$ value.

To create a repeated game scenario, both the attacker's and defender's current strategies are influenced by their past actions. On the defender's side, the No Security (NS) strategy is always chosen in the first round. This strategy is kept until an attack attempt is made, which causes the defender to choose the LS (Low Security) strategy. If the attack is successful, the defender chooses HS (High Security) for the next round. If there are a total of 2 successful attacks in the game history, the defender permanently chooses HS (High Security) for the remaining rounds of the game.

On the other hand, the attacker is initialized with the Basic Phishing (B) strategy. If their attack returns a positive payoff, their strategy is kept for the next round. Otherwise, the attacker chooses the Spear Phishing (S) strategy. If S returns a negative payoff, the attacker chooses N (No Attack), but if the payoff is positive, the attacker stays with S for the next round of the game. If the current round of the game is not N and the attacker's payoff has been negative for the last 3 or more rounds in which the attacker chose B or S, the attacker permanently chooses N for the remainder of the game. If none

of the conditions above are satisfied, the attacker chooses B for the next round.

V. METHODOLOGY

To analyze phishing email attacks using game theory, a repeated-game simulation was developed in Python to model strategic interaction between an attacker and a defender over multiple rounds. The simulation consisted of two players: the defender, representing an organization, and the attacker, representing a malicious adversary. Each player was given three strategies. The defender was given No Security (NS), Low Security (LS), and High Security (HS). The attacker was given No Attack (N), Basic Phishing (B), and Spear Phishing (S). In each round of the game, both players selected a strategy, utilities were computed for each player, and the outcome of the round influenced future decisions.

The program was implemented in Python using dictionaries, functions, and an iterative simulation loop. Dictionaries were used to store parameter values associated with each strategy, including security costs, attack costs, probabilities for successful attacks, attacker gains, losses, and penalties. The parameter values were chosen as placeholder values to represent real values, and were measured in millions. This structure allowed the payoffs to be computed for any combination of attacker and defender strategies. Functions were written to compute these components. One function calculated the probability of a successful phishing attack as the product of multiple stages in the phishing process, including delivery, user interaction, credential compromise, and monetization. Using this probability, utility functions for each player were constructed. The defender's utility was modeled as the negative of the total cost of security investment and expected losses from successful attacks. The attacker's utility was modeled as the expected gain minus attack costs and expected penalties from detection. Additional functions were created to compute defender losses, attacker gains, and the utilities generated by each strategy pair.

To make the game dynamic rather than static, the simulation incorporated a state variable representing defender awareness. Awareness increased after phishing attempts and decreased after periods of no attacks. This variable directly affected the probability of a user clicking a phishing link, which reduced attack effectiveness as awareness grew. In the Python program, awareness modified the click probability used in the success calculation, allowing the same strategy pair to produce different utilities over time. This models the effect of user training and learning, making the repeated game more realistic.

Strategic adaptation was implemented through additional decision functions that determined each player's strategy at the start of every round. The attacker updated behavior based on prior utility outcomes, continuing attacks when they are profitable and eventually choosing no attack when the utilities became negative. The defender adjusted security levels in response to observed attacks and successful breaches. By updating the strategies after each round, players are allowed to learn and adapt over time.

The game occurred in an iterative simulation loop that ran for 20 rounds. In each round, the program selected strategies for both players, computed success probabilities and utilities, determined randomly if an attack succeeded, updated awareness and strategic conditions, and stored the results in a history list. This list included strategies chosen, awareness levels, success probabilities, utilities, and attack outcomes for each round.

VI. RESULTS

The results produced by the Python program are shown in Figure 1. Initially, the attacker found phishing to be profitable as the defender had low awareness. In the first round, when the defender used NS (No Security), basic phishing yielded positive attacker utility, reflecting a favorable environment for the attack. After the defender transitioned to LS (Low Security), the attacker continued using phishing strategies, particularly spear phishing, because it initially produced positive utility despite repeated unsuccessful attack outcomes. As the game progressed, repeated phishing attempts caused the awareness to increase steadily from 0 to 5. This increase in awareness reduced the click probability used in the success model, causing the overall probability of a successful attack to be lower. As a result, the attacker's expected utility from spear phishing declined over time. This demonstrates the ability of increasing awareness to make attacks less profitable without requiring successful breaches to trigger escalated defense measures.

Though all of the attacks in this game resulted in unsuccessful outcomes, the attacker still adapted due to declining expected utility. Once the phishing strategies became unprof-

itable and resulted in negative utility, the attacker switched to the N (No attack) strategy and remained there for the remainder of the simulation. The game ultimately converged to the strategy pair (LS, N). This is significant because convergence was achieved without successful branches or escalation to HS (High Security). Rather than deterrence occurring from costly defense escalation, deterrence occurred through increasing awareness, as it reduced the effectiveness of attacks and the attacker's incentive to continue phishing attacks. This demonstrates that learning and awareness alone can alter incentives enough to stabilize the system.

Tables 1 and 2 show the payoff matrices of defender and attacker utilities at awareness levels 0 and 5, generated by the Python program simulation. The entries are given in the order of (UD, UA), representing the utilities of the defender and the attacker. The initial payoff matrix, Table 1, represents an environment with low awareness and higher phishing effectiveness. Spear phishing is profitable at this level under both defender strategies N (No attack) and LS (Low Security) since the payoffs are positive, while it is less profitable for HS (High Security) due to negative payoffs. The second payoff matrix, Table 2, shows how increasing awareness changes the incentives of the attacker.

Round	Defender	Attacker	Awareness	P(success)	Def Utility	Att Utility	Success
1	NS	B	0	0.224	-2.464	1.1498	False
2	LS	B	1	0.0525	-2.4987	-0.3759	False
3	LS	S	2	0.156	-5.12	0.708	False
4	LS	S	3	0.1365	-4.73	0.357	False
5	LS	S	4	0.117	-4.34	0.006	False
6	LS	S	5	0.0975	-3.95	-0.345	False
7	LS	N	5	0	-2	0	False
8	LS	B	5	0.0105	-2.0997	-0.6552	False
9	LS	S	5	0.0975	-3.95	-0.345	False
10	LS	N	5	0	-2	0	False
11	LS	N	5	0	-2	0	False
12	LS	N	4	0	-2	0	False
13	LS	N	3	0	-2	0	False
14	LS	N	2	0	-2	0	False
15	LS	N	1	0	-2	0	False
16	LS	N	0	0	-2	0	False
17	LS	N	0	0	-2	0	False
18	LS	N	0	0	-2	0	False
19	LS	N	0	0	-2	0	False
20	LS	N	0	0	-2	0	False
Total Defender Utility: -53.1525							
Total Attacker Utility: 0.4998							

Fig. 1: Output of the Python simulation.

As awareness increases, attacker utilities decrease across all strategies, with decreases being more significant for lower security levels. Both basic and spear phishing are less effective strategies at this awareness level, making N the most effective strategy to avoid negative payoffs at LS and HS. By comparing the two matrices, it is evident that increasing awareness shifts the game from favoring attacks to deterring attacks.

TABLE I: Payoff matrix at awareness level 0

	N	B	S
NS	(0, 0)	(-2.46, 1.15)	(-10.6, 7.8)
LS	(-2, 0)	(-2.6, -0.31)	(-5.9, 1.41)
HS	(-5, 0)	(-5.02, -1.01)	(-5.38, -2.51)

TABLE II: Payoff matrix at awareness level 5

	N	B	S
NS	(0, 0)	(-1.23, 0.29)	(-6.82, 4.39)
LS	(-2, 0)	(-2.1, -0.66)	(-3.95, -0.34)
HS	(-5, 0)	(-5.01, -1.02)	(-5.09, -2.76)

Figure 2 shows the utilities over time for each player. The attacker's utilities were mostly positive in the early stages of the game, but gradually became more negative as the game progressed. Convergence occurred when the attacker's utility stabilized at 0 due to deterrence. The defender utility remained negative throughout the simulation due to the costs of maintaining security and expected losses from attacks. However, it became less severe as the game progressed due to increasing defender awareness and decreasing success probabilities. Convergence occurred when the defender's utility stabilized at -2 due to maintaining LS. The ability of the defender utility to become less severe despite increasing costs and expected losses demonstrates the importance of awareness. Figure 3 displays the cumulative utility of each player over time. The defender's cumulative utility decreases over time due to ongoing security costs and exposure to phishing attempts, while the attacker's utility initially increases when phishing is profitable but eventually stabilizes at 0.

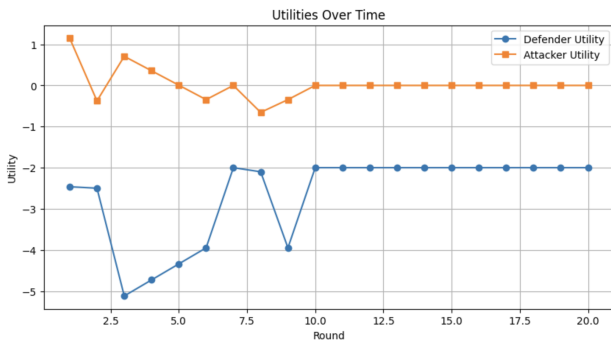


Fig. 2: Utilities over time for the attacker and defender.

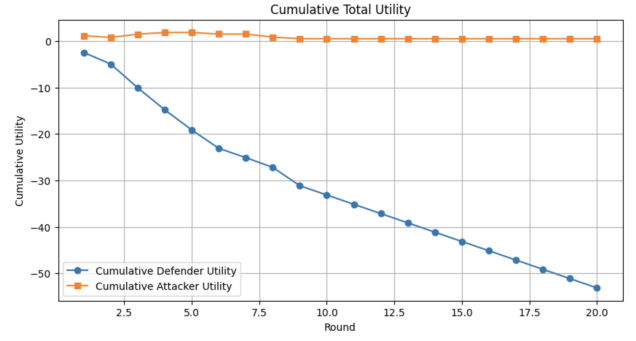


Fig. 3: Cumulative total utilities for both players.

Figure 4 displays the probability of success over time. At the start of the game, the probability of success was at its highest, at 22.4%. As the game progressed, it decreased and eventually converged to 0% when the attacker stopped attacking. Figure 5 displays the levels of awareness throughout the game. Awareness started at 0 but gradually rose to 5 by round 6 and remained there until round 12, where it decreased due to the attacker ceasing the phishing attempts. It can be observed through comparing the figures that increasing awareness decreases the probability of success, which leads to the decreased effectiveness of phishing attacks. These results highlight the role of awareness as a mechanism for deterrence, even in the absence of successful attacks or escalation to security.

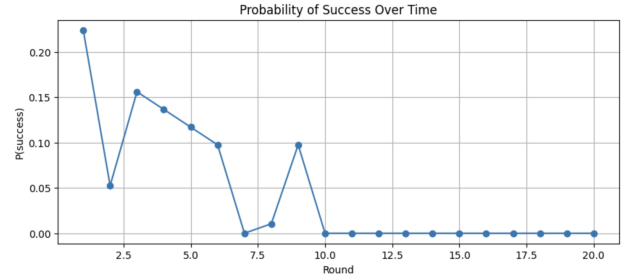


Fig. 4: Probability of successful phishing attacks over time.

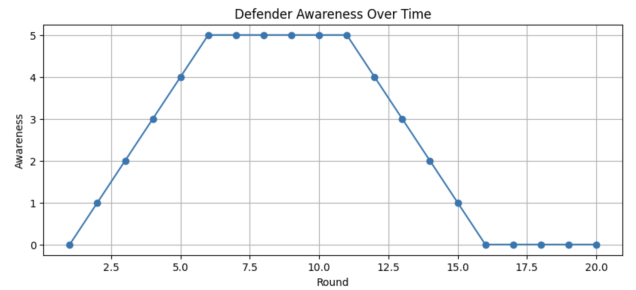


Fig. 5: Defender awareness over time during the simulation.

The results suggest that repeated-game modeling can help capture the dynamics in strategic decision-making. In particular, the simulation developed in this project shows that

awareness can alter attacker incentives over time, eventually making phishing attacks unprofitable even without successful breaches. Despite these findings, this model has several limitations. First, the parameters chosen for this simulation, such as probabilities, costs, and penalties, were chosen as representative placeholder values rather than estimated from empirical phishing data. While they were selected to be meaningful and consistent, different parameter choices could lead to different outcomes. Second, the model simplifies the attacker and defender behaviors. The strategy adaptation rules in this program are rule-based approximations rather than optimization or reinforcement learning. Additionally, there is only one dynamic factor being considered, which is awareness.

Finally, the simulation results are influenced by randomness in attack outcomes. Because attack success is determined using probabilities, different simulation runs may produce different paths to convergence.

VII. FUTURE WORK

Several improvements could be made to the model to better capture the complexity of real-world cybersecurity interactions. First, each player can have an increased number of strategies. The attacker could incorporate additional attack types, such as multi-stage phishing attacks or hybrid social engineering techniques, while the defender could include more nuanced responses, such as user training programs or incident response strategies. Expanding the available strategies would allow the model to capture a wider range of realistic scenarios and provide deeper insight into how different defensive measures impact attacker incentives.

Another way to improve the model is to introduce more dynamic factors beyond defender awareness. For example, an experience factor could allow attackers to increase their probability of success as more attacks are executed. This can give the defenders an incentive to strengthen their security early on in the game scenario. Additionally, the defender's behavior could be made more realistic by allowing decisions to be based on minimizing expected cost rather than following fixed rules.

The model has potential to be tested on real-world datasets rather than on generated placeholder values. With more dynamic factors and player strategies to simulate real-world cybersecurity scenarios, organizations can potentially use this model to help plan their defensive strategies against malicious phishing messages.

VIII. CONCLUSION

This study demonstrates how game theory can be applied to model cybersecurity interactions, specifically phishing email attacks, using a repeated-game simulation. By incorporating probabilities, utilities, and a dynamic, changing awareness level, the model was able to capture how attacker incentives and defender conditions evolve. The results showed that as awareness increases, the probability of successful attacks decreases, which in turn decreases attacker utility and leads to deterrence. The simulation eventually converged to a stable

outcome without successful breaches, highlighting the importance of defender learning and exposure in making strategic decisions. Overall, this study illustrates that cybersecurity outcomes are not determined solely by defensive strength, but also by how repeated interactions and dynamic factors like awareness influence long-term incentives.

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